

NovaTech AI - Technical Architecture Overview

1. System Architecture

NovaTech's platform is built on a cloud-native microservices architecture deployed on AWS. All services run in Kubernetes (EKS) clusters across three availability zones for high availability. The platform handles peak loads of 50,000 concurrent API requests with p99 latency under 200ms.

2. ML Infrastructure

The ML pipeline follows a standard train-evaluate-deploy cycle. Training uses PyTorch models on AWS SageMaker with spot instances achieving 70% cost savings. Experiments are tracked using MLflow on self-hosted EC2, currently tracking over 2,000 experiments. The model registry uses MLflow Model Registry with staging and production promotion workflows. Models are served via custom FastAPI inference servers containerized with Docker. Monitoring uses Evidently AI for data drift detection alongside Prometheus and Grafana for metrics.

3. Data Infrastructure

Real-time data is ingested via Apache Kafka (MSK on AWS) with 12 topics processing over 500GB of events daily. Batch pipelines run on Apache Airflow with 80+ active DAGs. The data warehouse is Snowflake, queried by the analytics team via dbt models.

The feature store uses Feast, backed by Redis for online features with sub-5ms reads and S3 for offline features. Feature definitions are version-controlled in Git and CI/CD pipelines automatically validate feature schemas on every pull request.

4. API Gateway and Security

All external API traffic flows through AWS API Gateway with rate limiting, request validation, and WAF rules. Internal service-to-service communication uses gRPC with mutual TLS. Secrets are managed by AWS Secrets Manager with automatic 90-day rotation. Authentication uses JWT tokens with 1-hour expiry. All access is logged to CloudTrail and analyzed by AWS GuardDuty for anomaly detection.

5. Disaster Recovery

Recovery Time Objective (RTO): 4 hours. Recovery Point Objective (RPO): 1 hour. Database backups run every hour to S3 with cross-region replication to eu-west-1. Full DR drills are conducted quarterly. The runbook is maintained on the internal portal.